

Shenglai Zeng

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RESEARCH INTEREST

I am a fourth-year Ph.D. candidate at Michigan State University, advised by Professor Jiliang Tang, and I am on the job market for positions starting in 2027. My research builds **personalized, proactive, and safe agents**. I work primarily on the **agent harness** — the self-improvement, memory, retrieval, and evaluation/environment machinery that determines what an agent can actually do — and extend it to **multimodal interaction** and **post-training**. I have published 35+ papers at NeurIPS, ICML, ICLR, ACL, EMNLP, NAACL, KDD, and CVPR, received the Best Paper Award of IEEE Transactions on Cloud Computing (2023) and an ACL-2026 SAC Highlight (top 1%, Best Paper Candidate), and have industry research experience at Meta AI, Amazon, and Baidu.

EDUCATION

Michigan State University

DSE Lab/Ph.D. Candidate in Computer Science and Engineering

East Lansing, U.S

Sept 2023-Present

Expected Graduation: May 2027

Advisor: Jiliang Tang

Lab: Data Science and Engineering Lab

Research Direction: Trustworthy AI, Large language models(LLMs)

University of Electronic Science and Technology of China

Yingcai Honor School/B.Sc in Computer Science and Engineering

Chengdu, China

Sept 2019 - Jun 2023

CGPA: 3.98/4.00

Weighted Average: 93.97/100(1st among 100 students)

Honors: The Most Outstanding Students Award of UESTC

HONORS AND AWARDS

- **SAC Highlight (Top 1%) & Best Paper Candidate**, ACL-2026 2026
- **Best Paper Award**, IEEE Transactions on Cloud Computing (JCR Q1) 2023
- **The Most Outstanding Students Award**, UESTC (highest honor; 10 students university-wide)
- **National Scholarship** (highest undergraduate honor in China) 2020
- WAC Scholarship (10 undergraduates university-wide per year), UESTC 2021
- 1st Outstanding Academic Scholarship (Top 5%), UESTC 2020, 2021, 2022

INDUSTRY EXPERIENCE

*All papers in this section are **first-authored** (including co-first).*

Meta AI

Research Scientist Intern/Next Generation Proactive Agent

Seattle, USA

Summer 2026

-Mentors: Yang Xiao, Luna Xin Dong, Scott Wen-tau Yih

- Memory-driven Proactive Assistance [**Submitted to ICLR 2027**].
- Collaborate with cross-organizational researchers spanning Meta Superintelligence Labs, FAIR, and Reality Labs on next-generation proactive multimodal agentic AI.

Amazon, Search & Shopping FM Team

Applied Scientist Intern/Retrieval-augmented Generation and Personalization

CA, USA

Summer 2024, 2025

-Mentor: Tianqi Zheng, Dante Everaert

-Manager: Hanqing Lu, Chuan Tian, Monica Xiao Cheng

- Efficient Personalized LLMs (PLLMs) [ACL-2026, Oral, Best Paper Candidate, SAC Highlight].
- Knowledge Checking in RAG [NAACL-2025, Oral].
- Representation Finetuning for Robust RAG [ACL-2025].
- Self-Improving Synthetic Data for RAG [EMNLP-2025].
- Multi-modal RAG [EMNLP-2025].

Baidu Inc., Search & Agent Business Team

Research Intern/RAG Privacy and LLM Memorization

Beijing, China

May 2023 - May 2024

-Mentor: Dr. Yiding Liu and Dr. Dawei Yin

- **First work to unveil the privacy risks of RAG** — 350+ citations [ACL-2024 Findings].
- Memorization in LLM post-training [ACL-2024].

PUBLICATIONS

35+ papers at top-tier ML/NLP venues (NeurIPS, ICML, ICLR, ACL, EMNLP, NAACL, KDD, CVPR), including **12 first-authored** publications. **1340+ citations** (Google Scholar, Sept. 2026). Recognitions include a **Best Paper Award** (IEEE TCC) and **Best Paper Candidate** (ACL), **Spotlight** (ICLR), **Oral** presentations (ACL, NAACL), and adoption in frontier-model technical reports (Muse Spark, Qwen, Kimi). * denotes equal contribution.

First-Authored Publications

- [NeurIPS'26] **S. Zeng**, Q. Wang, K. Guo, X. Dai, X. Long, H. Liu, "Magnifying What Matters: Attention-Guided Adaptive Rendering for Visual Text Comprehension", *Conference on Neural Information Processing Systems*, 2026.
- [ICML'26] **S. Zeng***, J. Zhang*, K. Guo, X. Dai, H. Liu, J. Tang, Y. Chang, "Fix Before Search: Benchmarking Agentic Query Visual Pre-processing in Multimodal Retrieval-Augmented Generation", *International Conference on Machine Learning*, 2026.
- [ACL'26] **S. Zeng**, T. Zheng, C. Tian, D. Everaert, Y.-S. Wang, Y. Huang, M. J. Morais, R. Patki, J. Tian, X. Dai, K. Guo, M. X. Cheng, H. Liu, "Attn-GS: Attention-Guided Context Compression for Efficient Personalized LLMs", *Annual Meeting of the Association for Computational Linguistics*, **Oral, SAC Highlight (Top 1%), Best Paper Candidate**, 2026.
- [EMNLP'25] **S. Zeng**, J. Zhang, P. He, J. Ren, T. Zheng, H. Lu, H. Xu, H. Liu, Y. Xing, J. Tang, "Mitigating the Privacy Issues in Retrieval-Augmented Generation (RAG) via Pure Synthetic Data", *Conference on Empirical Methods in Natural Language Processing*, 2025.
- [EMNLP'25] **S. Zeng***, J. Zhang*, J. Ren, T. Zheng, H. Liu, X. Tang, Y. Chang, "Beyond Text: Unveiling Privacy Vulnerabilities in Multi-modal Retrieval-Augmented Generation", *Conference on Empirical Methods in Natural Language Processing*, 2025.
- [ACL'25] **S. Zeng**, P. He, K. Guo, T. Zheng, H. Lu, Y. Xing, H. Liu, "Towards Context-Robust LLMs: A Gated Representation Fine-tuning Approach", *Annual Meeting of the Association for Computational Linguistics*, 2025.
- [NAACL'25] **S. Zeng**, J. Zhang, B. Li, Y. Lin, T. Zheng, D. Everaert, H. Lu, H. Liu, Y. Xing, M. X. Cheng, J. Tang, "Towards Knowledge Checking in Retrieval-Augmented Generation: A Representation Perspective", *Conference of the North American Chapter of the Association for Computational Linguistics*, **Oral**, 2025.
- [ACL'24] **S. Zeng***, Y. Li*, J. Ren, Y. Liu, H. Xu, P. He, Y. Xing, S. Wang, J. Tang, D. Yin, "Exploring Memorization in Fine-tuned Language Models", *Annual Meeting of the Association for Computational Linguistics*, 2024.

- [ACL'24] S. Zeng*, J. Zhang*, P. He, Y. Xing, Y. Liu, H. Xu, J. Ren, S. Wang, D. Yin, Y. Chang, J. Tang, "The Good and The Bad: Exploring Privacy Issues in Retrieval-Augmented Generation (RAG)", *Findings of the Association for Computational Linguistics: ACL*, 2024.
- [IEEE TCC] S. Zeng, Z. Li, H. Yu, Z. Zhang, L. Luo, B. Li, D. Niyato, "HFedMS: Heterogeneous Federated Learning with Memorable Data Semantics in Industrial Metaverse", *IEEE Transactions on Cloud Computing*, **Best Paper Award 2023**, 2023.
- [DASFAA'22] S. Zeng, Z. Li, H. Yu, Y. He, Z. Xu, D. Niyato, H. Yu, "Heterogeneous Federated Learning via Grouped Sequential-to-Parallel Training", *International Conference on Database Systems for Advanced Applications*, 2022.
- [SDM'23] S. Zeng*, J. Wang*, Z. Long, Y. Wang, H. Xiao, F. Ma, "Knowledge-Enhanced Semi-Supervised Federated Learning for Aggregating Heterogeneous Lightweight Clients in IoT", *SIAM International Conference on Data Mining*, 2023.

Co-Authored Publications

- [NeurIPS'26] Y. Li, Y. Feng, Z. Xu, Z. Ma, K. Zheng, F. Jiang, X. Sun, R. Shao, Z. Chen, Y. Huang, X. Han, B. Lee, K. Xu, S. Zeng, H. Hua, X. Zhang, B. Alomair, R. Krishna, L. Zettlemoyer, P. W. Koh, B. Ramasubramanian, L. Niu, X. Yue, R. Poovendran, "JobBench: Aligning Agent Work With Human Will", *Conference on Neural Information Processing Systems, Datasets & Benchmarks Track*, 2026. **(adopted by Meta Muse Spark 1.1, Moonshot Kimi K3, and Alibaba Qwen 3.8 Max)**
- [EMNLP'26] K. Guo, X. Dai, S. Zeng, H. Shomer, H. Han, Y. Wang, J. Tang, "Beyond Static Retrieval: Opportunities and Pitfalls of Iterative Retrieval in GraphRAG", *Conference on Empirical Methods in Natural Language Processing*, 2026.
- [ICML'26] X. Dai, K. Yang, C. Luo, S. Zeng, K. Guo, J. Tang, "When Do Hallucinations Arise? A Graph Perspective on the Evolution of Path Reuse and Path Compression", *International Conference on Machine Learning*, 2026.
- [KDD'26] K. Guo, X. Dai, Z. Zhang, N. Lin, S. Zeng, J. Ren, H. Han, J. Tang, "Why Retrieval-Augmented Generation Fails: A Graph Perspective", *ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2026.
- [NeurIPS'25] J. Ren, Z. Dai, X. Tang, Y. Xing, S. Zeng, H. Liu, J. Zeng, Q. Peng, S. Varshney, S. Wang, Q. He, C. C. Aggarwal, H. Liu, "Keeping an Eye on LLM Unlearning: The Hidden Risk and Remedy", *Conference on Neural Information Processing Systems*, 2025.
- [EMNLP'25] K. Guo, H. Shomer, S. Zeng, H. Han, Y. Wang, J. Tang, "Empowering GraphRAG with Knowledge Filtering and Integration", *Conference on Empirical Methods in Natural Language Processing*, 2025.
- [ACL'25] B. Wang, W. He, P. He, S. Zeng, Z. Xiang, Y. Xing, J. Tang, "Unveiling Privacy Risks in LLM agent memory", *Annual Meeting of the Association for Computational Linguistics*, 2025.
- [KDD'25] B. Li, Z. Chen, H. Han, S. Zeng, J. Liu, J. Tang, "Unveiling Mode Connectivity in Graph Neural Networks", *ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2025.
- [CVPR'25] J. Ren, K. Chen, Y. Cui, S. Zeng, H. Liu, Y. Xing, J. Tang, L. Lyu, "Six-CD: Benchmarking Concept Removals for Benign Text-to-Image Diffusion Models", *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.
- [TMLR] P. He, Y. Xing, H. Xu, J. Ren, Y. Cui, S. Zeng, J. Tang, M. Yamada, M. Sabokrou, "Stealthy Backdoor Attack via Confidence-driven Sampling", *Transactions on Machine Learning*

Research.

- [ECCV'24] J. Ren, Y. Li, **S. Zeng**, H. Xu, L. Lyu, Y. Xing, J. Tang, "Unveiling and Mitigating Memorization in Text-to-Image Diffusion Models through Cross Attention", *European Conference on Computer Vision*, 2024.
- [EMNLP'24] H. Xu, J. Ren, P. He, Y. Cui, **S. Zeng**, H. Liu, J. Tang, A. Liu, "On the Generalization of Training-based ChatGPT Detection Methods", *Findings of the Association for Computational Linguistics: EMNLP*, 2024.
- [ICLR'24] P. He, H. Xu, J. Ren, Y. Cui, **S. Zeng**, H. Liu, C. Aggarwal, J. Tang, "Sharpness-Aware Data Poisoning Attack", *International Conference on Learning Representations, **Spotlight***, 2024.
- [NeurIPS'23] J. Li, H. Shomer, H. Mao, **S. Zeng**, Y. Ma, N. Shah, J. Tang, D. Yin, "Evaluating Graph Neural Networks for Link Prediction: Current Pitfalls and New Benchmarking", *Conference on Neural Information Processing Systems, Datasets & Benchmarks Track*, 2023.

PREPRINTS

- [Preprint] X. Long, Z. Chen, **S. Zeng**, S. Wang, K. Guo, J. Tang, "MemTrace: Probing What Final Accuracy Misses in Long-Term Memory".
- [Preprint] X. Dai, C.-H. Lo, K. Guo, **S. Zeng**, D. Luo, J. Tang, "Uncovering Graph Reasoning in Decoder-only Transformers with Circuit Tracing".
- [Under review] Z. Chen, J. Gu, J. Yin, X. Long, **S. Zeng**, X. Liu, K. Guo, K. Zhou, J. Tang, "Exploring Cross-Scenario Generality of Agentic Memory Systems: Diagnostics and a Strong Baseline", *ACL Rolling Review*.
- [Under review] X. Dai, K. Guo, C.-H. Lo, **S. Zeng**, J. Ding, D. Luo, S. Mukherjee, J. Tang, "GraphGhosts: Tracing Reasoning Structures Behind Large Language Models", *ACL Rolling Review*.
- [Preprint] Y. Lin, P. He, H. Zhao, **S. Zeng**, Z. Xiang, H. Liu, J. Tang, "Structural Alignment Faking: Hiding Malicious Capabilities in Mixture-of-Experts LLMs".
- [Preprint] J. Ren, H. Xu, P. He, Y. Cui, **S. Zeng**, J. Zhang, H. Wen, J. Ding, H. Liu, Y. Chang, J. Tang, "Copyright Protection in Generative AI: A Technical Perspective".

Services

- **Conference Reviewer / Program Committee:** NeurIPS, ICML, ICLR; ACL, EMNLP, NAACL (via ACL Rolling Review); MICCAI Workshop on Distributed, Collaborative and Federated Learning (DeCaF-2022)
- **Journal Reviewer:** ACM Transactions on Information Systems (TOIS), IEEE TKDE, IEEE TMC, IEEE TPDS, IEEE Transactions on Information Forensics & Security, IEEE Transactions on Services Computing, IEEE Transactions on Vehicular Technology, IEEE Internet of Things Journal
- Reviewed **30+** papers.